

Probabilistic Relevance:

Vector space relevance feedback and probabilistic relevance feedback are used in information retrieval systems. It helps to increase the relevancy of the search engine. The query is given by the user for information retrieval, and using these techniques helps find the relevancy based on the query given by the user. As per it will give the results. They help to enhance the relevancy by better providing information. Thus, it gives the user satisfaction level for the result.

The differences between vector space relevance and probability are:

Vector space relevance feedback:

The vector space model is a mathematical framework that is highly used in information retrieval systems. Here, the document and the query are both in vector space. Then the search engine searches in the vector space model of queries given by the user and documents in high-dimensional space. In this way, it represents the relevant vector space model.

At first, the query is given by the user, and there are many documents, so as per that query, the relevant document will be retrieved. Following that, the user provides feedback based on whether or not the feedback is relevant. So, this feedback is just to determine whether the user is getting relevant or non-relevant results. Based on the feedback, it modifies the initial query vector given by the user to find the best match. This is done by calculating the weighted terms from the initial query vector. In this way, it finds the best relevancy from the revised query given by the users. Then it retrieves the best result as per that updated query. For example, it calculates the similarity between the queries given by the user and the document using cosine similarity techniques.

The several steps it includes are:

1. Document term frequency matrix. (DTM)
2. Term Frequency-Inverse Document Frequency (TF-IDF)
3. Vectorization
4. Cosine similarity

The documents are matched based on their relevance scores. Then the high-scoring document will be retrieved as the best match.

Feedback on Probabilistic Relevance:

Here, search engines use probabilistic models. They are the binary independence model or the language model. These help find the relevance of the documents. At first, the user gives the query, and it retrieves the relevant document based on that query. After that, the user gives feedback as per the relevant results of the document. (Ramishamukhtar, 2023) After that, it updates the probabilistic model based on the feedback given by the user, and then it evaluates the

likelihood terms as to what documents are relevant to it. Based on that input, it calculates or modifies the probabilistic model to retrieve the new or relevant result. Using these statistical models, it estimates relevance and updates the query accordingly. updating the query accordingly.

Several steps involved here are:

Let d-document

P (d/R, q)- Relevant document with respect to the given fixed query.

P (d/NR, q)- NR-non-relevant document with respect to the given query (fixed).

Now we have to find the P (R/d, q), which is the probability that document d is relevant to query q.

For this, we use the Bayes rule as below:

$$p(R | d, q) = \frac{p(d | R, q) p(R | q)}{p(d | q)}$$
$$p(NR | d, q) = \frac{p(d | NR, q) p(NR | q)}{p(d | q)}$$
$$p(R | d, q) + p(NR | d, q) = 1$$
$$O(R | d, q) = \frac{p(R | d, q)}{p(NR | d, q)}$$

Here, P(R/q) and P(NR/q) are the prior probabilities for retrieving relevant and non-relevant documents with query q. The total calculates the probability of each term in the document for relevance and the probability of that term in the document for non-relevance. Similarly, it calculates the same for the query as well. Then both probabilities are combined using the Bayes theorem for calculating the probabilities of the relevant.